

Pack a School Backpack SOP

One episode packs one backpack. The pack lies flat on its back on the table for the whole episode. Three books go into the main compartment flat, largest at the bottom, then the lunchbox on top of them, then the main zip closes. The pencil case goes into the front pocket and the front zip closes. The water bottle goes into the side pocket last. Every staging spot is empty when the episode ends.

The order never changes: books, lunchbox, main zip, pencil case, front zip, bottle. A compartment is closed before the next one is opened.

Every move is the same shape. The left gripper holds the pack, either gaping a mouth open or pressing the pack down for a zip. The right gripper makes every pick, every insertion, and every zip draw. The left arm never carries an item and the right arm never holds the pack.

The two grippers never work the same end of the pack. The left gripper stays at the pack's left end. Every pick, every insertion, and every draw happens to the right of it.

The pack never leaves the spot it was set on, nothing is turned in the air, and no gripper opens anywhere but over the compartment the item is going into.

Setup

Complete both checklists before starting an episode.

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole tabletop: the pack lying flat at the back, and the book row, lunchbox spot, case spot, and bottle spot down the right.
3. The whole top face of the pack is in frame: both zip tracks end to end, and the right side pocket.
4. The main compartment mouth is visible from above, so every insertion can be seen landing.
5. Both arms are at home with grippers open.
6. The tabletop is clear of anything but the zones listed below.
7. Nothing obstructs the paths between the staging spots and the pack, and the right arm reaches the left end of both zip tracks while the left arm rests on the pack's left end.

Materials checklist

1. One school backpack lying flat on the table at the back, back panel down on the tabletop and front panel facing up, squared to the table with the top of the pack toward the back edge and the base toward the front edge.

2. Both compartments are empty and unzipped, each slider parked at the **left end stop** of its track.
3. Both zips are straight tracks running left to right across the top face of the pack, each with a single slider carrying a fabric loop tab big enough for a gripper to pinch.
4. The right side pocket is empty and holds its shape with nothing in it, its mouth open and facing the back edge, so a bottle can be slid straight in.
5. Three books of visibly different size lying flat in the book row, front covers up, largest nearest the pack, side by side with a clear gap between them so a gripper can close on one book without touching its neighbors.
6. One rigid lunchbox at the lunchbox spot, lid closed and facing up, sitting flat on its base.
7. One rigid box shaped pencil case lying flat at the case spot, zipped shut, long enough to be grasped across its two long sides.
8. One rigid water bottle lying on its side at the bottle spot, cap closed, base pointing toward the pack, resting where it was set without rolling.
9. Every staging spot holds one item and nothing rests on top of anything else.

Workspace layout

- **Pack spot:** at the back of the tabletop, centered between the two arms and set close enough that both arms reach the whole top face of the pack without extending. The pack lies flat on its back here and stays there for the whole episode.
- **Book row:** at the back right, three books lying flat side by side, largest nearest the pack. It is the only place a book is picked up from.
- **Lunchbox spot:** on the right, inboard of the case spot, within reach of the right arm.
- **Case spot:** on the right, outboard of the lunchbox spot.
- **Bottle spot:** at the front right, within reach of the right arm.
- **Left side of the tabletop:** kept clear. The left arm's whole working area is the pack, approached from the left. Nothing is ever staged or set down there.

The staging spots run in the order the items are used, nearest the pack first, so a carried item never passes over another item.

Vocabulary

- **Arm assignment:** the left arm holds. It gapes each mouth and presses the pack down for each zip. The right arm makes every pick, every insertion, and every zip draw. Neither arm takes the other's job, and no item ever touches the left gripper.
- **Main compartment:** the large compartment under the main zip. It takes the three books and the lunchbox.
- **Front pocket:** the flat pocket on the front panel, which faces up while the pack lies flat. It takes the pencil case and nothing else.

- **Side pocket:** the open pocket on the pack's right side. It takes the bottle and nothing else, and it has no zip.
- **Back panel:** the padded face of the pack resting on the tabletop. While the pack lies flat it is the floor of the main compartment, and the books stack up from it.
- **Mouth:** the opening of a compartment while it is held open.
- **Gape:** the left gripper pinching the **left end** of a mouth's front lip and lifting it up and toward the front edge, so the mouth stands open as a slot facing up. A gape is held for the whole step, not pulled and released.
- **Track:** the line of teeth a slider runs along. It has a **left end stop**, where the slider parks while the compartment is open, and a **right end stop**, where it parks when closed.
- **Loop tab:** the fabric loop on a slider. It is the only part of a zip a gripper ever touches.
- **Anchor:** the closed left gripper pressed down on the pack's **left end**, clear of both tracks, holding the pack flat and square on the table while the right gripper draws a zip. An anchor never travels, and it never sits on a track.
- **Draw:** the right gripper pinching the loop tab and pulling the slider along the track, left end stop to right end stop, in one continuous motion along the line of the track.
- **Load:** the right gripper pinching an item, lifting it clear of its spot, carrying it over the open mouth, lowering it straight down until it is resting, and releasing.
- **Books:** named the **largest**, the **medium**, and the **smallest**, never by title or color.
- **Settled:** the item stays put for 2 seconds after the gripper lifts clear, with nothing rocking, sliding, or rolling.
- **Release point:** over the compartment or pocket the item is going into, once the item is already resting on its floor. A gripper opens nowhere else.

Steps

Steps 1 to 6 are one backpack. Every item is taken straight down onto its grasp, lifted straight up until there is daylight under it, and carried in the orientation it will be loaded in.

Step 1: Load the three books, largest first

Goal: three books lying flat and stacked in the main compartment, largest at the bottom, stepping down in size to the top.

- The **left gripper** gapes the main compartment: it pinches the **left end of the mouth's front lip** and lifts it up and toward the front edge until the mouth stands open, and holds it there for the whole step.
- The **right gripper** pinches the **largest** book across its two long edges, spine on one side and open edge on the other, lifts it straight up clear of the book row, and carries it flat and level.

- Lower it flat through the **right side of the mouth** until it lies on the back panel, and release once it is resting.
- Repeat with the **medium** book, then the **smallest**, each one lowered flat onto the one before it. Nothing else changes between the three.

Gape before the book is picked up. A mouth that has fallen shut takes the book onto the closed top face instead of into the compartment.

Check: three books lie flat and stacked in the compartment, largest at the bottom and smallest on top, roughly squared to each other, and the stack sits below the mouth line. If a book is on edge, out of size order, or hanging over the mouth, lift it clear of the pack and lower it in again.

Step 2: Load the lunchbox

Goal: the lunchbox flat on top of the book stack, lid up.

- The **left gripper** keeps the gape from Step 1. It does not move in this step.
- The **right gripper** pinches the lunchbox across its two short sides, lifts it straight up, and carries it to the pack lid up and level.
- Lower it through the **right side of the mouth** onto the top book until it rests flat. Release once it is resting.
- Both grippers lift clear and the mouth falls shut on its own.

Check: the lunchbox sits flat and level on top of all three books, lid up, with no part of it standing above the mouth line, and the books are still stacked in size order. If it is on an angle, bridging the mouth, or has pushed a book out of the stack, lift it clear of the pack and lower it in again.

Step 3: Zip the main compartment shut

Goal: the main zip closed end stop to end stop, with nothing caught in it.

- The **left gripper**, closed, anchors the pack: it presses down on the pack's **left end**, clear of both tracks, and holds it there for the whole step.
- The **right gripper** comes down onto the **loop tab** from above the track, where the slider is parked at the left end stop, and pinches it.
- Draw the slider straight along the track, left to right, in one continuous motion, until it reaches the right end stop and will not travel further. Release the tab and lift clear.
- The **left gripper** lifts clear only once the right gripper is off the track.

The draw pulls the pack rightward across the table, and the anchor is the only thing keeping it square. The anchor stays at the left end so the two grippers never meet over a track.

Check: the slider sits at the right end stop, the track is joined along its whole length, and no fabric, strap, book corner, or lunchbox edge is caught in the teeth. If the slider stopped short,

take the loop tab again and draw it the rest of the way.

Step 4: Load the pencil case into the front pocket

Goal: the pencil case fully inside the front pocket.

- The **left gripper** gapes the front pocket: it pinches the **left end of the pocket mouth's front lip** and lifts it up and toward the front edge until the mouth stands open, and holds it there.
- The **right gripper** pinches the pencil case across its two long sides, lifts it straight up, and carries it to the pack flat and level.
- Lower it flat through the **right side of the pocket mouth** until it rests on the pocket floor, and release.
- Both grippers lift clear and the pocket falls shut.

The main compartment is closed first because a slack top face gives the pocket mouth nothing to open against.

Check: the pencil case is fully inside with no part of it at the mouth. If the case stands proud of the mouth, lift it out, gape the pocket again, and lower it back in.

Step 5: Zip the front pocket shut

Goal: the front zip closed end stop to end stop, with nothing caught in it.

- The **left gripper**, closed, anchors the pack on its **left end** as in Step 3, and holds it there for the whole step.
- The **right gripper** comes down onto the **loop tab** from above the front pocket track and pinches it.
- Draw the slider left to right in one continuous motion to the right end stop. Release the tab and lift clear, then the left gripper lifts clear.

Check: the slider sits at the right end stop, the track is joined along its whole length, and nothing is caught in the teeth. If the slider stopped short, take the loop tab again and draw it the rest of the way.

Step 6: Load the water bottle into the side pocket

Goal: the bottle lying in the side pocket, base at the bottom of the pocket and the cap end at the mouth.

- The **left gripper**, closed, presses down on the pack's **left end** and holds the pack square on the table for the whole step.
- The **right gripper** pinches the bottle around its body below the cap, lifts it straight up, and carries it lying on its side with the base pointing at the pocket mouth.

- Line it up with the pocket and slide it in base first, in one straight motion along the pocket, until the base reaches the bottom of the pocket and the bottle stops. Release with the cap end still at the mouth.
- Both grippers lift clear.

Check: the bottle lies in the side pocket, base in and cap end at the mouth, with most of its length inside the pocket, and settled. The pack is still square on the table. If the bottle sits across the mouth, hangs more out than in, or has pushed the pack out of square, lift it clear and slide it in again.

Step 7: End the episode

- Confirm both sliders sit at their right end stops, the bottle lies in the side pocket, and every staging spot is empty.
- Return both arms home, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Slide the bottle out of the side pocket and lay it on its side at the bottle spot, base pointing toward the pack.
2. Draw both sliders back to their left end stops and leave both compartments open.
3. Lift the pencil case out of the front pocket and lay it flat at the case spot.
4. Lift the lunchbox and the three books out of the main compartment. Set the lunchbox flat at the lunchbox spot, lid up.
5. Lay the three books flat in the book row, front covers up, largest nearest the pack, side by side with a clear gap between them.
6. Square the pack on the table: back panel down, front panel up, top of the pack toward the back edge. Confirm it is empty.
7. Check both loop tabs are attached and both zips run freely end to end. A zip that is starting to catch is replaced before the next episode.
8. Replace any item that is damaged: a side pocket that has lost its shape, a book whose cover is falling off, a lunchbox whose lid falls open, or a leaking bottle.
9. Wipe the tabletop and confirm nothing has been left in the pack spot or the staging spots.
10. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Wrong order

- **Visible cue:** the lunchbox goes in before all three books, the main zip is drawn before the lunchbox is in, the front pocket is opened before the main compartment is closed, or the bottle goes in before the front zip is drawn.
- **SOP rule broken:** Steps 1 to 6, the order never changes and a compartment is closed before the next one is opened.
- **Coaching note:** finish the compartment you are in before you open another.

Violation: Arms swapped roles

- **Visible cue:** the left gripper picks up, carries, or loads an item, or draws a zip; the right gripper gapes a mouth or anchors the pack while the left works.
- **SOP rule broken:** Steps 1 to 6, the left arm holds and the right arm works, and no item ever touches the left gripper.
- **Coaching note:** left holds, right works.

Violation: Both grippers on the same end of the pack

- **Visible cue:** during a zip the left gripper sits on or beside a track instead of at the pack's left end, or the two grippers crowd the same end of a track.
- **SOP rule broken:** Steps 3, 5, and 6, the anchor presses down on the pack's left end clear of both tracks, and the two grippers never work the same end of the pack.
- **Coaching note:** the anchor belongs at the left end, never on the zip line.

Violation: Item fetched from the left

- **Visible cue:** the right gripper travels to the left of the pack to pick something up, or an item is staged or set down on the left side of the tabletop.
- **SOP rule broken:** Steps 1 to 6, every item is staged on the right and the left side is kept clear.
- **Coaching note:** everything the right arm needs is on its own side.

Violation: Loaded without a hold

- **Visible cue:** the right gripper brings an item to a compartment whose mouth is shut, sets it down on the closed top face, or the left gripper takes the gape only after the item is in the air or comes off it before the item is resting.
- **SOP rule broken:** Steps 1, 2, and 4, the mouth is held open before the item is picked up and stays held for the whole step.
- **Coaching note:** gape first, let go last.

Violation: Zip drawn without an anchor

- **Visible cue:** the left gripper is off the pack while a zip is drawn, or the pack slides, twists, or is dragged across the table as the slider travels.
- **SOP rule broken:** Steps 3 and 5, the pack is anchored at its left end for the whole draw.
- **Coaching note:** anchor first, then pull.

Violation: Pack lifted or moved

- **Visible cue:** the pack is lifted off the table, dragged, rotated, stood up, or turned over at any point, including being pulled out of square by a gape.
- **SOP rule broken:** Steps 1 to 6, the pack stays flat on its back where it was set.
- **Coaching note:** the gape lifts the lip, not the pack. Nothing here needs the pack picked up.

Violation: Item taken by the wrong part

- **Visible cue:** a book is taken by one cover, by a corner, or by its pages; the lunchbox by its lid or handle; the pencil case by its zip; the bottle by its cap; or a zip is taken by the slider body, the teeth, the tape, or the fabric beside it rather than by the loop tab.
- **SOP rule broken:** Steps 1 to 6, each item has one grasp and the loop tab is the only part of a zip a gripper touches.
- **Coaching note:** land on the grasp point or reset and take it again.

Violation: Item dragged instead of lifted

- **Visible cue:** an item slides along the tabletop still touching it, with no daylight under it in transit.
- **SOP rule broken:** Steps 1 to 6, lift straight up until there is daylight before moving horizontally.
- **Coaching note:** the lift is its own motion, not the start of the carry.

Violation: Item turned in the air

- **Visible cue:** a book is tipped up on edge or turned over during the carry, the lunchbox is tilted off level, the pencil case is turned on its edge, or the bottle is tipped upright or end over end between its spot and the pack.

- **SOP rule broken:** Steps 1 to 6, nothing is turned in the air and each item is carried in the orientation it will be loaded in.
- **Coaching note:** orientation is set at the pick, never in the air.

Violation: Regripped in the air

- **Visible cue:** the gripper shifts, rolls, or seats its hold again on an item without setting it down first.
- **SOP rule broken:** Steps 1 to 6, set it down before taking a new hold.
- **Coaching note:** put it down before you take a new hold.

Violation: Book stack wrong

- **Visible cue:** the medium or smallest book goes in before the largest, the finished stack does not step down in size from the bottom up, or a book is lowered in on edge, leaning against a wall of the compartment, or crossways with a corner over the mouth.
- **SOP rule broken:** Step 1, largest first flat on the back panel, then medium, then smallest, each lying flat on the one before it.
- **Coaching note:** the row is already in loading order. Work it from the pack outward and lay each book down flat.

Violation: Wrong compartment

- **Visible cue:** the pencil case goes into the main compartment, a book or the lunchbox goes into the front pocket, or the bottle goes anywhere but the side pocket.
- **SOP rule broken:** Steps 1, 2, 4, and 6, each item has one compartment.
- **Coaching note:** one compartment per item, fixed, every episode.

Violation: Item dropped instead of lowered

- **Visible cue:** a gripper opens above a mouth and the item falls the rest of the way, or an item is angled or pushed in on edge rather than lowered flat until it is resting.
- **SOP rule broken:** Steps 1, 2, and 4, release only once the item is resting.
- **Coaching note:** take it all the way down before you open the gripper.

Violation: Gripper released over the wrong place

- **Visible cue:** a gripper opens over the tabletop, over a staging spot, over the closed pack, or over open space, and the item lands anywhere but inside a compartment.
- **SOP rule broken:** Steps 1 to 6, a gripper opens over the compartment the item is going into and nowhere else.
- **Coaching note:** know where it lands before you open the gripper.

Violation: Contents pressed or packed down

- **Visible cue:** a gripper presses, shoves, or squashes items already in a compartment to make room, or pushes the book stack down to fit the lunchbox.
- **SOP rule broken:** Steps 1 and 2, each item is lowered onto the last one and nothing is packed down.
- **Coaching note:** if it does not fit, that is data. Forcing it is not.

Violation: Zip drawn off the track or backwards

- **Visible cue:** the slider is pulled up off the pack, lifted off the fabric, drawn in an arc rather than along the teeth, or drawn right to left, and the track bunches, gaps, or the slider stalls.
- **SOP rule broken:** Steps 3 and 5, one continuous motion along the line of the track, left end stop to right end stop.
- **Coaching note:** keep the tab moving along the teeth, not away from them.

Violation: Zip left short of the end stop

- **Visible cue:** a draw finishes with the slider short of the right end stop and an open gap in the track, and the episode moves on; or the episode ends with a compartment unzipped or partly zipped.
- **SOP rule broken:** Steps 3, 5, and 7, both sliders sit at their right end stops.
- **Coaching note:** run it to the stop, and look at the stop before you move on.

Violation: Something caught in the zip

- **Visible cue:** fabric, a strap, a book corner, the lunchbox edge, or the pencil case is trapped in the teeth as the slider passes, and the draw continues anyway.
- **SOP rule broken:** Steps 3 and 5, nothing is caught in the teeth and no item stands above the mouth line.
- **Coaching note:** contents standing proud of the mouth is what causes this. Check the load sits low before you draw.

Violation: Bottle not seated in the pocket

- **Visible cue:** the bottle is carried end up, pushed in cap first, laid across the mouth of the pocket, left hanging more out than in, or set on top of the pack.
- **SOP rule broken:** Step 6, carry it on its side and slide it in base first until the base reaches the bottom of the pocket.
- **Coaching note:** base leads the way in, and it goes all the way in.

Violation: Placement corrected by shoving

- **Visible cue:** a gripper nudges, pats, or slides an item, or the pack, into position after releasing it, instead of lifting it clear and lowering it again.

- **SOP rule broken:** Steps 1 to 6, correct a bad placement by lifting the item clear of the pack and lowering it in again.
- **Coaching note:** lift it and lay it again, or leave it alone.

Violation: Item dropped, collided, or knocked something over

- **Visible cue:** an item falls onto the tabletop or the floor, the pack is struck and shifts, the book row is scattered, a staged item is displaced, or the two arms strike each other.
- **SOP rule broken:** Steps 1 to 6, keep each item on a clear travel path and release only after a settled placement.
- **Coaching note:** check the path and the landing spot before you move.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with an item still on a staging spot, a zip not at its right end stop, the bottle not in the side pocket, or an arm away from home.
- **SOP rule broken:** Step 7, confirm both zips, the bottle, and the empty staging spots; return both arms home; then stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Zip jammed or slider derailed** under a straight draw along the track. Replace the pack before the next episode.
- **Loop tab missing or torn off** when the draw begins.
- **Defective item:** a side pocket that will not hold the bottle, a book whose cover is falling off, a lunchbox whose lid falls open, or a leaking bottle. Replace it before the next episode.
- **The items will not all fit** in the compartments they are assigned to.

Annotation subtasks (from SOP)

1. Hold a compartment open
2. Move one book from the book row into the main compartment
3. Move the lunchbox into the main compartment
4. Anchor the pack for a zip
5. Draw a zip closed

6. Move the pencil case into the front pocket
7. Slide the water bottle into the side pocket
8. Return both arms home and end the episode